# Annotation

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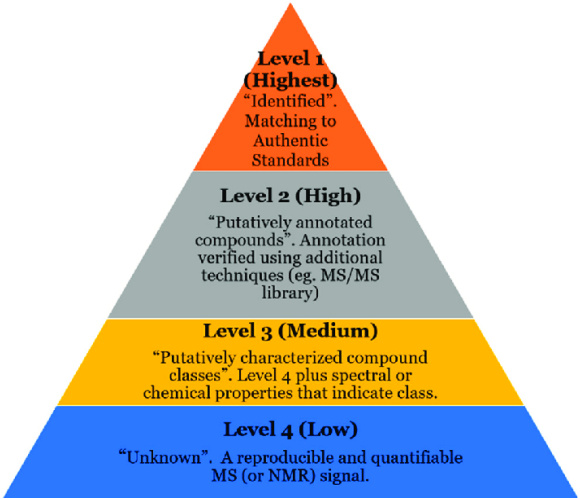
# Annotation

Annotation is the process of **going "back"** from the features measured in an untargeted metabolomics experiment to the corresponding metabolites.

## Challenges

1. Unknown metabolites
2. Poor standardization of chromatography (in particular LC)
3. Bioinformatic infrastructures (*databases*,*workflows*, ...)

# Annotation Levels

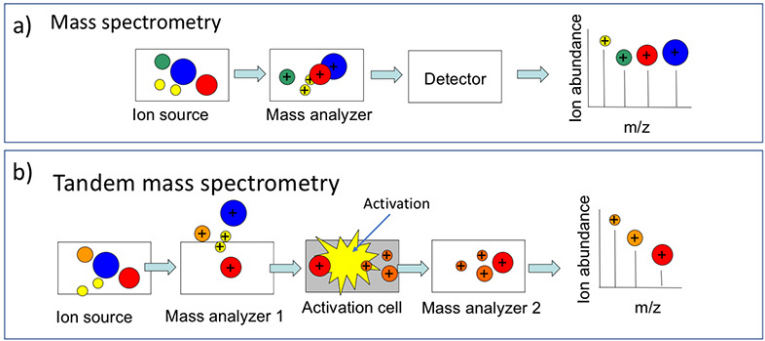


# The tools for the trade

In the case of MS based experiments, the process of annotation requires the interpretation of the ionization pattern of each molecule and the additional information which can be derived from chromatography and fragmentation experiments

1. Chemical expertise
2. Database of standards (in-house or on-line)
3. Fragmentation experiments
4. Molecular Networks
5. Chemoinformatics and predictive models

# Fragmentation Experiments




- Fragmentation patterns are sensitive to the chemical structure
- MS/MS can be matched with databases of standards

# Databases of Standards

- Buy a chemical standard, analyze it with your instrument (MS, MS-MS, retention time)
- PubChem
- MassBank
- Human Metabolome Database
- Meltin
- ...

# Chemoinformatics and predictive models

- Sirius
- Metfrag

# Molecular Networks

- GNPS